Consumption, Saving, and Investment

Lecture Notes*

Ravikiran Naik • Department of Economics • FLAME University

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Learning objectives.

1. Describe the factors that affect consumption and saving decisions (§4-1).
2. Discuss the factors that affect the investment of firms (§4-2).
3. Explain the factors affecting goods market equilibrium (§4-3).

§	Topic
4-1	Consumption and Saving
4-2	Investment
4-3	Goods Market Equilibrium

Table 4.1: Sections covered.

- **This chapter builds the demand side of the goods market.** Chapter 3 fixed the real *supply* side — full-employment output $\bar{Y}$ from labour and the production function. Here we ask how that output is *used*: desired consumption C^d, desired investment I^d, and the real interest rate that makes desired spending equal output.

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- **Keep the desired-vs-actual distinction in front of you.** Equilibrium is about *planned* spending; the accounting identity $Y = C + I + G$ always holds because unsold goods are counted as inventory investment. The equilibrium condition can fail; the identity cannot.
- **The whole section pays off in one picture — the S–I diagram.** The real interest rate adjusts until desired national saving equals desired investment. Everything before it (how C^d, S^d, I^d, and K^* are determined) is machinery for reading that diagram.
- **Consumption is the largest piece.** It is over two-thirds of spending, and investment — though only about one-sixth — is the most volatile piece and the engine of long-run growth. Between them they drive the business cycle (see the appendix).

4.1 Consumption and Saving

4.1.1 The Importance of Consumption and Saving

- **Where we are.** Chapter 3 gave us output *supplied*; this chapter turns to the **demand** for goods — how that production is actually *used*. Aggregate demand has **four parts**: C (consumption), I (investment), G (government purchases), and NX (net exports). We simplify by taking G as **given** (set by the political process) and assuming a **closed economy** so $NX = 0$, which leaves the focus on C and I.
- **Desired consumption** ⇒ what households *plan* to consume (not actual)
  - **Why start with consumption?** It is **over two-thirds** of total US spending — the largest single demand component — so shifts in households' willingness to spend move the whole economy.
  - **Flag the desired-vs-actual distinction now:** it recurs all chapter, because goods-market equilibrium will require $Y = C^d + I^d + G$, where each term is the *desired* level.
  - **Measurement red flag.** The national accounts (NIPA) count spending on durables when it happens; theory wants only the *services* a durable yields over time. The two diverge, so measured consumption is not exactly the consumption the model has in mind.
  - **The deeper link.** Studying the demand for goods *is* studying **saving and capital formation**: saving is simply what is left after the consume decision, and investment *is* the purchase of new capital.
- **Desired national saving** ⇒ national saving when C is at its desired level
  - The counterpart equation: once C is at its desired level C^d, national saving is whatever output remains after private consumption and government purchases.

$$S^d = Y - C^d - G \tag{4.1}$$

4.1.2 The Consumption and Saving Decision of an Individual

- **Text example — Prudence.** She earns $60,000 after taxes. She could consume the whole $60k, but she has two other options, and the choice between them is really a single decision about how to split income between now and later.
- Consume **less** than income ⇒ saving **positive**

  - **To save** is to consume less than \$60k and accumulate the difference for later use — retirement being the classic real-world saving motive.
- Consume **more** than income ⇒ saving **negative** (dissaving)
  - **To dissave** is to spend more than income by running down wealth or borrowing. If Prudence borrows \$5000 she consumes \$65k now, but she must then consume *less* than income later to repay — dissaving today forces saving tomorrow.
  - **One decision, not two.** Since saving is just income minus consumption, choosing how much to consume *is* choosing how much to save. Evelyn earns \$8000; if she consumes \$7400 she automatically saves \$600. Decide either figure and the other follows — two sides of the same coin.
  - **Keep the terms distinct.** *Saving* is a flow (what is not consumed this period); *savings* is a stock (wealth accumulated over time). The chapter is about the flow.
- **Two theories stand behind this chapter.** The forward-looking view comes in two complementary forms: Friedman's **permanent-income hypothesis** (1957) is about how people respond to *random* income receipts, while the Modigliani–Brumberg **life-cycle hypothesis** (1954) is about *predictable* income over the life cycle — low saving when young, peak saving around ages 50–60, dissaving in retirement. They are complementary, not rival.

4.1.3 Current versus Future Consumption

- **Current ⇆ future consumption is a trade-off.** A household can always move consumption across time by saving or borrowing, and the real interest rate sets the terms on which it can do so.
- Price of 1 unit today ⇒ $1 + r$ units of future consumption (r = real interest rate)
  - **Compounding (forward):** multiply by $(1 + r)$. At $r = 10\%$, Rs 100 saved today becomes Rs 110 next year.
  - **Discounting (backward):** divide by $(1 + r)$. Rs 110 next year is worth Rs 100 today.
  - **So the "price" of one unit today is $(1 + r)$ units forgone in the future.** Skip saving Rs 100 and you give up Rs 110, so consuming now costs 1.10 in future terms. The real interest rate r *is* the price of the trade-off.
- **Consumption-smoothing motive** ⇒ prefer an *even* consumption path over time
  - **Definition.** The desire for a relatively even pattern of consumption across time, avoiding spells of very high or very low consumption.
  - **Why both extremes are rejected.** Borrow heavily now and you have almost nothing to consume later; save almost everything and you near-starve today. A household chooses an interior path instead — it spreads a one-off windfall over years and cushions an income dip by saving or borrowing.
  - **This motive drives the rest of the section.** It is *why* the income, future-income, and wealth effects below take the shapes they do (with the real interest rate held fixed for each).
  - **A trade-off students already live.** The clearest present-vs-future choice most of them face is *being in college*: forgoing pay now (often borrowing) to raise future income, with the interest rate as the price of trading present for future.

4.1.4 Effect of Changes in Current Income

- Current income ↑ ⇒ consumption **and** saving both ↑ (↓ ⇒ both fall)
  - **The response is symmetric:** a fall in current income lowers *both* C and S.
  - **Ask why MPC < 1.** Households save part of any extra income precisely to smooth future consumption, so not all of a raise is spent.
  - **Permanent vs temporary (Friedman's permanent-income theory).** A *temporary* rise (y ↑ with y^f unchanged) barely moves lifetime resources, so it is **mostly saved**; a *permanent* rise (both y and y^f up) raises lifetime resources much more, so it is **mostly consumed**.
- **MPC** = share of extra income consumed ⇒ $0 < MPC < 1$
  - Part of any extra income is consumed and the rest saved, which pins the MPC strictly between 0 and 1.
  - **Worked example — \$6000 bonus, MPC = 0.4.** Consumption rises $0.4 \times 6000 = \$2400$ and saving rises the remaining \$3600 — she neither splurges it all nor banks it all.
  - **Symmetric layoff, MPC = 0.4.** Income falls \$8000, so consumption falls \$3200 and saving falls \$4800.
- Aggregate: income ↑ ⇒ C^d ↑ (by less than Y) ⇒ S^d ↑
  - Because C rises by less than Y, the remainder of the extra income flows straight into national saving.
- **Expected future income** ↑ ⇒ consumption ↑ today ⇒ saving ↓
  - **Smoothing pulls future income forward:** a household expecting a raise consumes more now and saves less.
  - **The mechanics.** Current income is unchanged, so a Rs 1000 rise in current consumption *is* a Rs 1000 fall in current saving. That lower saving cuts next year's resources by Rs $1000 \times (1 + r)$ — but smoothing means the household uses the expected raise to lift consumption both now and later.
  - This channel is the basis of the permanent-income (Friedman) and life-cycle (Modigliani) theories.

4.1.5 The Keynesian Consumption Function

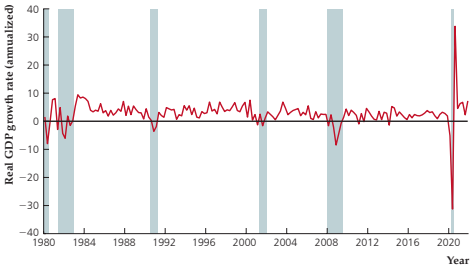
$$C^d = c_0 + c_y Y$$

- **A simple rule of thumb:** consume a base amount plus a fixed fraction of income. The slope c_y is the MPC and the intercept c_0 is autonomous consumption. This contrasts with the forward-looking permanent-income and life-cycle theories.
- C^d = total consumption spending
- c_0 = autonomous consumption (consumption when income is zero)
  - This is the floor the function predicts households would still consume even at zero income — financed, implicitly, by dissaving.
- c_y = marginal propensity to consume (MPC), where $0 < c_y < 1$
  - Here the MPC is the *slope* of the straight-line function, so it is assumed constant — a convenient

simplification rather than a behavioural claim.

- Y = disposable income
- **The cross-section data behind the Keynesian form.** US Consumer Expenditure Survey deciles show the *average* propensity to consume falling as income rises — the lowest decile spends about \$23,600 on \$6,800 of after-tax income (APC far above 1, i.e. dissaving), the middle decile \$44,000 on \$42,400 (APC ≈ 1), the top decile \$137,000 on \$205,000 (APC ≈ 0.67). A positive intercept with a slope below one — exactly $c_0 + c_y Y$ (Geerolf, ECON 102, from CEX).
- **Theories of Consumption** The Keynesian form is a *rule of thumb*; the three forward-looking theories instead agree that consumption tracks *lifetime*, not current, income — each solving the "consumption puzzle" (why the APC falls across households, yet is flat over the long run, as the decile data above and Kuznets both show). They differ in what drives the income gap and how policy bites.
  - **Life-cycle (Modigliani, 1954)** — income varies *predictably* over life (work, then retire), so $C = \alpha W + \beta Y$: consumption depends on **wealth as well as income**. Save when young, dissave when old; because wealth shifts the intercept αW up over time, the long-run APC stays flat even as the short-run function looks Keynesian.
  - **Permanent-income (Friedman, 1957)** — income = permanent + transitory ($Y = Y^P + Y^T$), so $C = \alpha Y^P$: consumers **spend permanent income and save most of transitory income**. The low measured MPC out of *current* income is an *errors-in-variables* artifact. A *temporary* tax change (the 1968 surcharge) barely moved C; a *permanent* one (the 1964 cut) did.
  - **Random-walk (Hall, 1978)** — permanent-income *plus* rational expectations ⇒ changes in consumption are **unpredictable**: only *news* (surprises) about lifetime income moves C, so only *unanticipated* policy has an effect. **Caveat:** in the data a predictable \$1 fall in income still cuts consumption by about \$0.50 ("excess sensitivity"), pointing to liquidity constraints.

4.1.6 Application: The Pandemic Recession and Consumption



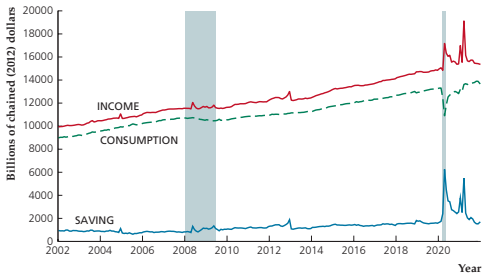
Real GDP around the 2020 pandemic recession: Q2 2020 was the largest quarterly decline on record and Q3 2020 the largest increase, yet the recession lasted only two months — the shortest ever. FRED real GDP series `GDPC1`.

Take-away. Even the sharpest output collapse can be brief when the shock itself (not a shortfall of demand) is doing the work.

- A record collapse and a record rebound — the shortest recession ever
  - **Timing (Fig. 4.1).** Q2 2020 saw the biggest GDP decline in history, immediately followed by the largest increase in history in Q3 2020.
  - **Depth and length.** The drop was deeper than the 2008 crisis and the worst since the Great Depression, yet the recession was the *shortest* on record, lasting only two months.
  - **The policy twist.** Governments *replaced* lost incomes through expanded unemployment insurance and direct cash, so after-tax disposable income actually **rose** during the recession.
  - **Why saving jumped.** Even with income up, consumer spending fell sharply as people avoided contact — travel, dining — and the gap between higher income and lower spending

showed up as a surge in saving.

4.1.7 Application: Income, Consumption, and Saving



Monthly income, consumption, and saving, Jan 2002 – Dec 2021. Income spiked on government programs while consumption dropped, so saving rose substantially for over a year.

Take-away. When income is propped up but spending is blocked, the gap between the two *is* the surge in saving.

- Income spiked, consumption fell, and saving rose for over a year
  - **What Fig. 4.2 plots:** monthly income, consumption, and saving from January 2002 through December 2021.
  - **The pandemic pattern.** Income spiked on government programs while consumption dropped on contact avoidance, pushing saving up substantially for more than a year.
  - **A composition shift, not just a level shift.** Spending moved from *services* toward *goods*, which changed the demand for workers across sectors.
  - **Open question.** Whether these shifts prove temporary or permanent is not yet settled.

4.1.8 Effect of Changes in Wealth

- Wealth ↑ ⇒ current consumption ↑ ⇒ current saving ↓
  - **Recall the definition (Ch 2):** wealth = assets − liabilities.
  - **Text example — Prudence's stock certificate.** She finds 50 forgotten shares worth $6000. For her resources this has the same effect as a $6000 income rise.
  - **But she raises consumption by *less* than $6000**, saving the rest for future consumption. Since income is unchanged, the extra current consumption equals an equal **fall** in current saving.
  - **Symmetric:** a fall in wealth makes her consume less and save more.
  - **Where wealth changes come from.** Stock-market ups and downs are a major source — the hook for the later Application on the boom and bust.
- **The wealth effect is real but empirically weak.** On 19 Oct 1987 the Wilshire 5000 fell 18% in a day, erasing about $1 trillion — yet consumption fell less than $40 billion and no recession followed. In the 2008 crisis US household net wealth dropped nearly 20% ($66.7T → $55.0T), but the consumption/GDP ratio fell less than one percentage point. Households spread a wealth change over the whole lifetime, so the marginal propensity to consume out of wealth is *small* (Abel Ch. 4).
- **Saving is concentrated at the top.** On Saez–Zucman data the bottom 90% save close to zero (often negative) while the top 1% save 30–60% of income — so aggregate saving is driven disproportionately by the wealthy, a caveat to reading S^d off one representative household (Geerolf, ECON 102).

4.1.9 Effect of Changes in the Real Interest Rate

- **A rise in the real interest rate sets off two opposing effects.** This is the hardest idea in the chapter, so it pays to separate the substitution effect from the income effect and work them one at a time.
- **Substitution effect** ⇒ bigger reward for waiting ⇒ saving ↑
  - **Why.** Each unit saved now grows to $(1 + r)$ next year, so a higher r raises the *price* of current consumption. Consumers substitute away from now-pricier present consumption toward the future, and current saving rises.
- **Income effect** — works in opposite directions for savers and borrowers
  - **Saver.** A higher r grows her saving faster, so she needs to set aside *less* to hit a fixed goal (the \$1400-laptop case) — she consumes more now and saves less. A saver also earns more interest income at higher r, which acts like a wealth gain and lifts both current and future consumption, again cutting saving. (The Rs 110 target is worked out numerically on the Worked Example slide below.)
  - **Borrower.** A rate rise is a *wealth loss* — they are locked in on worse terms — so they rebuild wealth by saving more. For a borrower the income effect pushes saving *up*, the same direction as the substitution effect.
- Empirics: mixed ⇒ probably a *slight* ↑ in aggregate saving
  - **Why national saving is ambiguous.** The economy mixes savers (whose two effects offset) and borrowers (whose effects both raise saving), so theory alone cannot sign the net result — we must look at data.
  - **What the data say.** Evidence is mixed; the most-accepted reading is that a higher r reduces consumption and raises saving, but the effect is *not strong*.
  - **IRA example.** An Individual Retirement Account shelters saving from tax up to an annual cap. *Concretely:* if the cap is \$5,000, your first \$5,000 of saving earns the higher sheltered return — the steeper budget segment (substitution effect, save more) — while saving beyond \$5,000 reverts to the ordinary taxed return, kinking the line back. But the sheltered return also makes you richer (income effect, save less), so the two kinks pull opposite ways and the net is ambiguous. Empirically each \$100 of IRA saving cuts current consumption by about \$32.
  - **The telling exception.** For someone who would have saved *more* than the IRA contribution limit even without it, the account delivers *only* an income effect (richer, but no change in the marginal reward to the next dollar saved) — so for that person an IRA actually *lowers* saving. The incentive bites only on savers at or below the limit.

4.1.10 Worked Example: A Rate Rise — Saver vs Borrower

- **Saver**, targeting Rs 110 next year — income effect ⇒ saves **less**
  - At $r = 10\%$ she needs $110/1.10 =$ Rs 100 today; at $r = 20\%$ she needs only $110/1.20 \approx$ Rs 92. Same goal, higher rate, so she saves less.
  - **Picture a bucket being filled to a target line:** a faster tap means you reach the line with less water poured in.

- **Borrower**, owing at the old rate — wealth loss ⇒ saves **more**
  - The rate on what they owe rises (say 5% → 10%), so the old terms are now unfavourable — a wealth loss.
  - To rebuild wealth they consume less and save more. Netted against the substitution effect across the whole economy, the result is small and slightly positive.

4.1.11 Taxes and the Real Return to Saving

$$r_{a\text{-}t} = (1 - t)i - \pi^e \qquad (4.2)$$

- Expected after-tax real interest rate
  - **Reading (4.2).** You earn a nominal rate i, tax takes a share t so you keep $(1-t)i$, and subtracting expected inflation π^e leaves the real return.
  - **Non-numeric point.** Tax shrinks the reward for waiting, so it *blunts* the substitution effect that would otherwise raise saving.
- Calculating after-tax rates, with $i = 5\%$ and $\pi^e = 2\%$
  - At $t = 30\%$: after-tax nominal $= (1 - 0.30)\,5\% = 3.5\%$, real $= 1.5\%$. At $t = 20\%$: after-tax nominal $= (1 - 0.20)\,5\% = 4\%$, real $= 2.0\%$ (Abel's Table 4.1).
  - **Why this rate.** The after-tax real rate measures the true rise in purchasing power that saving buys, so it is the *right* rate for the consumption and saving decision. Lower tax on interest raises the after-tax return and strengthens the saving incentive.
  - **Policy — IRAs.** These shelter interest from tax to raise the after-tax return and stimulate saving; but because the interest-rate effect on saving is weak and disputed, their effectiveness is *also* disputed.
  - **Sheshinski (Data App).** Because *nominal* rather than real returns are taxed, US after-tax real rates were negative in the 1970s. In 1985 a pretax 6.3% became an after-tax 0.9% ($\pi = 3.6\%$, $t = 55\%$); in 1987 a pretax 4.9% became 2.1% ($\pi = 3.7\%$, $t = 33\%$).
  - **Classroom example.** Invest \$10,000 at $i = 5\%$, $\pi^e = 2\%$, $t = 30\%$: the nominal return is \$500, of which inflation eats \$200 and the government takes \$150, leaving only \$150 in real terms — a *real* tax rate of 50%. In the double-digit-inflation 1970s–80s the real tax rate on interest income ran well over 100%.

4.1.12 In Touch with Data and Research: Interest Rates

In Touch with Data & Research: Interest Rates

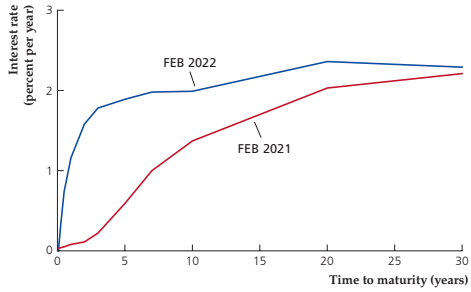
Many different interest rates exist (the Fed's H.15 release lists prime, federal funds, 3- and 6-month T-bills, 10-year notes, 30-year bonds), yet macro models usually speak of a single "the" interest rate. The resolution is that rates differ for a few systematic reasons but move together over time. A snapshot (H.15, 23 Feb 2022) shows both the spread and the co-movement: prime 3.25%, federal funds 0.08%, 3-month T-bill 0.36%, 10-year note 1.99%, 30-year bond 2.29% — a year earlier every one of these was lower, with short rates having risen the most.

- Rates differ by ⇒ **default risk**, **term** (yield curve), **tax status**

- **Know the names.** The *prime rate* is what banks charge their best customers; the *federal funds rate* is what banks charge each other on overnight loans; Treasuries are US-government debt.
- **Default risk.** Government debt is treated as risk-free, and riskier borrowers must pay extra, which is why prime and federal-funds rates sit above the risk-free rate.

- Rates co-move ⇒ we say "**the**" interest rate
  - Because the various rates move up and down together, models can collapse them into one "the" interest rate without losing much.
- **Yield curve** ⇒ a bond's term (life) vs. the rate it pays
  - This plots the rate against maturity for *same-risk* bonds. Longer-maturity bonds usually pay more, so the yield curve generally slopes upward. The full theory — the term structure of interest rates — comes in Ch 7.

4.1.13 The Yield Curve



The yield curve: yield to maturity against term to maturity, holding risk fixed (same issuer). It usually slopes upward (normal); a flat curve signals transition, and an inverted curve — short rates above long — often precedes a recession.

Take-away. The *shape* reads the cycle: upward is normal, inverted (short > long) is the classic recession warning.

- Plots yield against term to maturity, risk held fixed
  - **What it plots.** Yield (yield to maturity) against term to maturity, holding risk fixed by using one issuer — G-secs in India, Treasuries in the US. The x-axis runs from 3 months to 30 years; the y-axis is the annualised yield.
  - **Reading the shape.** *Upward* (long > short) is normal; *flat* signals transition or uncertainty; *inverted* (short > long) signals expected rate cuts and often appears before a recession.
  - **Default risk, separately.** The chance of non-repayment shifts a whole curve: higher risk means a higher rate demanded at every maturity.

4.1.14 Fiscal Policy

- Affects C^d ⇒ via current & expected future income
  - Government purchases and taxes change households' current and expected future income, which shifts C^d.
  - **Simplifying assumption.** Take output Y as given (full employment, Part 2), so fiscal changes here shift *demand*, not supply.
  - **The rule of thumb.** Any change that raises the private tax burden, now or in the expected future, makes people consume less.
- Affects S^d **directly** ⇒ the $-G$ term in $S^d = Y - C^d - G$

- G enters desired saving *directly* through the $-G$ term, a channel separate from and on top of its effect through consumption.

4.1.15 Government Purchases (Temporary Increase)

- $G \uparrow$ + current taxes $\uparrow \Rightarrow$ after-tax income $\downarrow \Rightarrow C^d \downarrow$
  - **Text example.** G rises \$10 bn (military) financed by \$10 bn of current taxes, so after-tax income falls \$10 bn and C^d falls — say by \$6 bn.
- Same if financed by *future* taxes ⇒ provided people foresee them
  - The outcome is the same whether taxed now or later, *so long as* households foresee the future tax bill and respond to it today.
- C^d falls *less* than G rises ⇒ both C^d **and** $S^d \downarrow$
  - **The numbers.** C^d down \$6 bn *raises* S^d by \$6 bn, but G up \$10 bn *lowers* it, so net S^d falls \$4 bn (from $S^d = Y - C^d - G$ with Y fixed). Consumption falls by *less* than G rises, so national saving falls too.
  - **Evidence.** Ahmed (1986) finds for the UK that temporary increases in G crowd out consumption, consistent with this prediction.
- **Temporary vs permanent G — a preview of the S–I diagram.** Only a *temporary* rise in G shifts the saving curve (consumers cut C by less than ΔG, so $S^d = Y - C^d - G$ falls, pushing r up). A *permanent* equal rise in current *and* future G makes forward-looking consumers cut C by roughly the full ΔG, so desired saving barely moves and the market-clearing r is unchanged. The interest-rate effect comes from the *temporary* component.

4.1.16 Taxes

- Lump-sum tax cut now ⇒ financed by higher *future* taxes
  - This is a pure *timing* shift of taxes — the total burden is unchanged.
  - **Text setup.** Cut T by \$10 bn lump-sum (about \$100 each to 100 million taxpayers) with G held constant, so the government must borrow \$10 bn now and repay it with higher future taxes.
- Future income $\downarrow$ offsets current income $\uparrow \Rightarrow C^d$ could rise **or** fall
  - Current income is up \$10 bn, which pushes C up (by less than \$10 bn); but expected future income falls as future taxes loom, which pushes C down. If the future loss exactly equals the current gain there is no change in C — an ambiguity that leads straight to Ricardian equivalence.

4.1.17 Ricardian Equivalence

- Future loss = current gain (exact) ⇒ **no change** in consumption
- Tax change ⇒ only *timing*, not total (same present value)
  - **The key insight is present value.** What matters is the *total* tax burden over time, not when it is collected; the current gain and the future loss have the same present value and exactly cancel, leaving C and S^d unchanged. The modern proposition is due to **Barro (1974)**; it is *named* for David Ricardo, who raised the logic but doubted it in practice (O'Driscoll, 1977, calls this the "Ricardian *non*equivalence" theorem).

- **Accounting check.** Private saving rises \$10 bn while government saving *falls* \$10 bn (a bigger deficit), so **national saving is unchanged**.

- In practice: future taxes *not foreseen* ⇒ cut feels like income ⇒ $C^d \uparrow$, $S^d \downarrow$
  - If households do *not* foresee the future taxes, the cut feels like extra income, so C rises and S^d falls.
  - **Why it may fail.** Liquidity constraints; finite horizons unless people leave bequests (Barro 1974); the cut going to different people than those who repay it; distortionary taxes; and uncertainty. The classic review is Bernheim (1987).
  - **Rebate evidence.** After the 2001 rebate households spent little extra (Shapiro–Slemrod), but after 2008–09 they spent 50–90% (Parker et al.) — both against strict Ricardian equivalence.
  - **Credit-card microdata (Agarwal, Liu & Souleles 2007).** Tracking the 2001 rebate on card accounts, households at first *paid down* balances (saving), but within nine months had *raised* spending and run their balances back up. The response was sharply uneven: **younger, borrowing-constrained** households spent the most, while those with **high credit limits** paid down and spent little — behaving most like Ricardian consumers. Liquidity constraints, not pure myopia, drive the departures.
  - **A good question to pose:** who really pays for a deficit-financed tax cut? Future taxpayers, in present-value terms.
  - **The real burden, traced out.** If saving *does* fall (future taxes unforeseen), the chain is $S^d \downarrow \Rightarrow r \uparrow \Rightarrow I \downarrow \Rightarrow$ a **smaller future capital stock**. The debt's true cost is not the repayment itself but the lost capital — future generations inherit a *less productive* economy.

4.1.18 Worked Example: A Tax Cut Today, Higher Taxes Tomorrow

- **Setup:** a Rs 100 lump-sum tax cut now, repaid via higher taxes next year, with $r = 10\%$.
- Future tax bill: Rs 100×1.10 = Rs 110 next year
  - The government borrows Rs 100 to fund the cut and must repay 100×1.10 = Rs 110 next year through higher future taxes.
- Present value of that bill: Rs $110/1.10$ = Rs 100
  - Discounting the Rs 110 back one year gives $110/1.10$ = Rs 100 — exactly the size of the original cut.
- Change in lifetime resources: Rs 100 − Rs 100 = 0
  - Adding +Rs 100 today and −Rs 100 (the present value of the future tax) leaves lifetime resources *unchanged*.
- ⇒ A forward-looking household saves the whole cut; C and S^d unchanged
  - **Intuition.** The government gave nothing away — the bill still comes — so the household simply banks the cut, and national saving is unchanged.
  - **Where it breaks.** If the future tax is not foreseen (myopia), the household treats the cut as income, so C rises and S^d falls.

4.2 Investment

4.2.1 Why Is Investment Important?

- The **most volatile** component of GDP ⇒ drives the **business cycle**
  - **Small share, outsized swings.** Investment is only about **one-sixth of GDP**, yet in a typical recession **half or more** of the total ↓ in spending is falling investment. That is why a chapter on the goods market opens with investment: it is the part that moves.
  - **Two time horizons, one variable.** Investment matters for short-run **fluctuations** (this chapter) *and* for long-run **growth** through capital accumulation (Ch. 6). The same spending flow does both jobs, which is why it keeps reappearing in the course.
- Builds the capital stock ⇒ drives long-run growth
  - Every unit of investment that survives depreciation becomes next period's capital. Consumption is enjoyed and gone; investment is the only component of demand that also enlarges future supply — the link back to the production function of Ch. 3.

4.2.2 The Desired Capital Stock

- **Desired capital stock** K^* ⇒ capital that *maximises* expected profit
  - **Read "desired" as "optimal".** K^* is not what the firm *has*; it is the stock it *would choose* if it could set capital freely. Investment is then just the effort to move the actual stock toward K^* — which is why we pin down K^* first and only afterward ask how fast firms get there.
- Set by ⇒ marginal **benefit** vs marginal **cost** of more capital
  - This is the *same* optimizing logic as the labour-demand rule $MPN = w$ from Ch. 3, now applied to capital. Add capital while the extra output it brings beats the cost of holding it; stop where they are equal.
- Investing pays off with a *lag* ⇒ benefit = **future** MPK (MPK^f)
  - **Why the "f" superscript.** Capital bought today produces *next* period, so the relevant benefit is the **expected future** marginal product, MPK^f, not today's MPK. The firm must forecast, and that forward-looking character is what later makes the stock market (q) and expectations matter.
  - **Comparative statics to keep in your pocket:** a higher expected MPK^f → **higher** K^*; a higher user cost → **lower** K^*; a higher effective tax rate → **lower** K^*. All three run through the same channel — they change profit per extra unit of capital.
- **What counts as "capital" is shifting.** Abel's homogeneous K increasingly means *intangibles*: software and R&D (intellectual-property products) rose from about 5% of GDP in the net capital stock in 1938 to roughly 23% by 2018. Tech firms hold little physical capital, so the MPK^f-vs-user-cost logic now applies to intangibles whose depreciation and measurement differ sharply from machines (Geerolf, ECON 102, from BEA).

4.2.3 Example: Kyle's Bakery

- **Kyle's Bakery** ⇒ buys a **solar-powered oven**, choosing its size
  - **The textbook's running example.** Kyle's Bakery, a maker of specialty cookies, is weighing a new oven to bake more cookies in future — and must also decide *how big* to build it. That size choice is the whole point: it turns "invest or not" into "how much capital."
- Price ⇒ **$100 per cubic foot** (real, any size, continuously divisible)
  - Because the oven can be *any* size, capital is treated as continuously divisible, so we can talk about the *marginal* cubic foot — exactly what the $MPK^f = uc$ rule needs. Price $p_K = \$100$ per cubic foot.
  - Being solar powered, the oven has **no energy or maintenance costs**; the only decay is productive (fewer cookies each year). That is deliberate — it strips the example down to interest and depreciation alone.
- Depreciation ⇒ **10% fewer cookies/yr** → real value falls **10%/yr** (→ $90)
  - Here depreciation is **physical/productive decay** (fewer cookies), which maps cleanly into a 10% fall in real value per year: $d = 10\%$. A $100 cubic foot is worth $90 after one year.
- Finance ⇒ borrow or lend at real rate **8% per year**
  - **Borrow-or-lend at the same r is doing real work.** It makes the **opportunity cost** of self-financing equal to the borrowing cost: either way, a cubic foot of oven ties up $100 that could have earned 8% elsewhere. $r = 8\%$.
  - **These three numbers** — $p_K = \$100$, $d = 10\%$, $r = 8\%$ — feed straight into the user-cost formula on the next slide. Memorise the setup; the arithmetic follows.

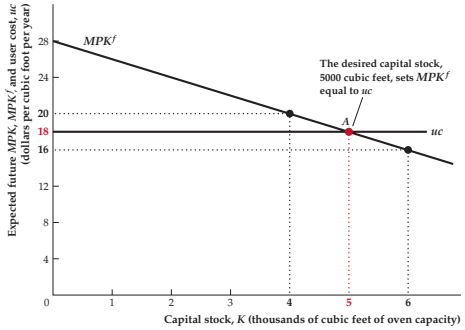
4.2.4 The User Cost of Capital

- Kyle's Bakery ⇒ price p_K, depreciation d, real rate r
  - Carry the numbers forward: $p_K = \$100$ per cubic foot, $r = 8\%$, $d = 10\%$.
- **User cost** = cost of using 1 unit for a period ⇒ interest cost + depreciation

$$uc = rp_K + dp_K = (r + d)p_K \qquad (4.3)$$

- Think of uc as the **rent** on one unit of capital for a year, whether or not the firm literally rents it: **interest forgone** plus **wear-and-tear**.
- **Worked numbers (Kyle).** Interest $rp_K = 0.08 \times 100 = \$8$; depreciation $dp_K = 0.10 \times 100 = \$10$. So user cost $uc = (0.18)(100) = \$18$ per cubic foot per year — 18% of the oven's value.
- **Red flag students miss:** even *self-financed* capital has an opportunity cost. The $8 of forgone interest is real whether Kyle borrows the $100 or pays cash; spending your own money is not free.
- The user-cost concept is due to **Jorgenson (1963)**; it is the foundation for the whole desired-capital analysis that follows.

4.2.5 Determination of the Desired Capital Stock



Desired K sits where the downward-sloping MPK^f curve crosses the flat uc line, at point A — here $K^* = 5000$ cubic feet, where $MPK^f = uc = \$18$.

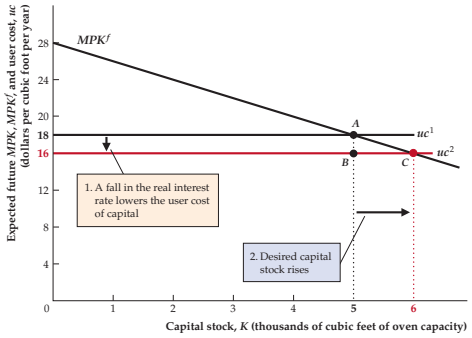
Take-away. Optimal capital is where marginal *benefit* (MPK^f) meets marginal *cost* (uc) — the capital version of $MPN = w$.

- **Desired capital is where $MPK^f = uc$.** It is marginal benefit = marginal cost for capital — the identical optimizing logic as labour ($MPN = w$), now read off two curves.
- **Why the curves have these shapes. MPK^f falls** as K rises (diminishing returns to capital); uc is **flat** in K (renting one more cubic foot costs the same \$18 regardless of how much you already hold).
- **How the firm reasons to the optimum.** Where $MPK^f > uc$ (e.g. $K = 4000$, $MPK^f = \$20$), add capital — it pays more than it costs. Where $MPK^f < uc$ (e.g. $K = 6000$, $MPK^f = \$16$), cut capital. Profit is maximised only where $MPK^f = uc$, at $K^* = 5000$ cubic feet (point A).

4.2.6 Factors That Change the Desired Capital Stock

- Desired capital sits where $MPK^f = uc$, so it moves *only* when one side of that equation moves.
- **User cost uc changes** ⇒ via r, depreciation d, or price p_K
- **MPK^f curve shifts** ⇒ via technology (capital more or less productive)
  - **Same logic, two doors.** Anything that changes the marginal **benefit** (MPK^f) *or* the marginal **cost** (uc) will move K^*; nothing else can. The next two figures take one door each — **Fig. 4.4** is a change in uc, **Fig. 4.5** is a shift in MPK^f.

4.2.7 A Decline in the Real Interest Rate Raises the Desired Capital Stock



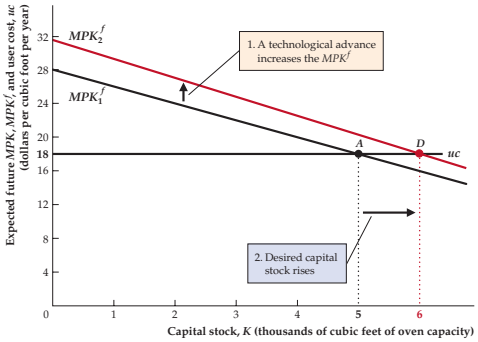
A lower r drops the user-cost line ($uc_1 \to uc_2$); the crossing with the unchanged MPK^f curve slides right, raising desired capital from A to C.

Take-away. Anything that lowers the user cost raises desired capital — you slide *along* a fixed MPK^f curve.

- **Worked numbers (Kyle).** The real rate falls r: **8% → 6%**, so user cost falls uc: **\$18 → \$16** and the uc line shifts down ($uc_1 \to uc_2$). Desired capital rises from **5000 (point A) to 6000 (point C)**, where $MPK^f = uc = \$16$.
- **The intuition behind the mechanism.** A cheaper interest cost makes holding capital less burdensome, so the firm can profitably hold *more* of it before the falling MPK^f meets the lower

uc. This is the micro-foundation of the downward-sloping investment-demand response to r that drives the goods-market diagram later in the chapter.

4.2.8 An Increase in the Expected MPK^f Raises the Desired Capital Stock



A technological advance shifts the whole MPK^f curve up ($MPK_1^f \to MPK_2^f$); with *uc* unchanged, the crossing moves right, raising desired capital from A to D.

Take-away. A productivity gain raises desired capital by *shifting* the benefit curve, not by moving along it — contrast with Fig. 4.4.

- **Worked numbers (Kyle).** Kyle invents faster-baking dough (+12.5% cookies), so the MPK^f curve shifts **up 12.5% at every** K ($MPK_1^f \to MPK_2^f$). User cost is unchanged at \$18, so desired capital rises from **5000 (point A) to 6000 (point D)**, where $MPK^f = uc = \$18$ again.
- **The general rule.** With *uc* fixed, a higher expected MPK^f **raises** the desired capital stock. Note the contrast with the previous figure: a fall in r moves the *cost* line, a productivity gain moves the *benefit* curve, but both push K^* the same direction (right).

4.2.9 Taxes and the Desired Capital Stock

- With taxes ⇒ firm keeps only $(1-\tau)\,MPK^f$
  - The tax takes a share τ of capital's return, so only the after-tax piece is the firm's to compare against cost.
  - **Worked example.** Kyle pays **20%** of revenue in tax, so a \$20 pre-tax MPK^f yields only **\$16** after tax (\$4 to the government). The firm must compare the *after-tax* \$16 to the user cost, not the headline \$20.
- Set after-tax return = user cost ⇒ $(1-\tau)MPK^f = uc$ ⇒

$$MPK^f = \frac{uc}{1-\tau} = \frac{(r+d)p_K}{1-\tau} \qquad (4.4)$$

- **Gist** ⇒ the tax is a **wedge** on the return: the firm must earn a higher *pre-tax* MPK^f to clear the same user cost (since $\frac{1}{1-\tau} > 1$) ⇒ desired K **falls**
  - **The logic, step by step.** The no-tax rule is $MPK^f = uc$. A tax τ on the return leaves $(1-\tau)MPK^f = uc$, i.e. $MPK^f = uc/(1-\tau)$. Since $1/(1-\tau) > 1$, the required *pre-tax* MPK^f rises; with diminishing MPK that means desired K^* **falls**. In short, the tax discourages investment.
  - **Numbers (Kyle).** With $uc = \$18$ and $\tau = 20\%$, the firm must now earn pre-tax $MPK^f = 18/(1-0.20) = 18/0.80 =$ **\$22.50** to clear the same \$18 user cost — up from \$18 untaxed — so it holds less capital.
  - **The $(1-\tau)$ factor sets the size of the wedge.** The whole tax effect runs through $\frac{1}{1-\tau}$. A **higher** τ shrinks $(1-\tau)$, enlarging the required pre-tax return ($\tau = 40\% \Rightarrow 18/0.60 =$ **\$30**).

Depreciation allowances and investment tax credits, by pushing the effective τ below the statutory rate, shrink the wedge and **raise** K^*.

- **Why assume a clean revenue tax?** Revenue is observable and no accounting rules are needed, so the key result shows up with a single parameter τ.
- **Why that is unrealistic.** Real tax codes tax **profit, not revenue**. Depreciation allowances cut taxable profit (smaller bill, higher K^*); investment tax credits act like a lower p_K (lower user cost, higher K^*); interest is often deductible too.
- **The fix — an effective tax rate.** Define τ_{eff} as the revenue-tax rate that would give the *same* K^* as the real code, so we keep writing $MPK^f = uc/(1-\tau_{\text{eff}})$. In practice τ_{eff} is usually *below* the statutory rate (even negative with large credits).

4.2.10 Taxes and the Desired Capital Stock (continued)

- **Tax-adjusted user cost** $\Rightarrow \dfrac{uc}{1-\tau}$
  - A higher τ raises the **effective cost** of capital — the firm's hurdle rate — even though the physical user cost uc has not changed.
- $\tau \uparrow \Rightarrow$ tax-adjusted user cost $\uparrow \Rightarrow$ desired $K \downarrow$
  - In practice the many features of a real tax code are collapsed into a single **effective tax rate**, with depreciation allowances and investment tax credits netting out against the statutory rate. The textbook's **Summary Table 4.2** reports effective rates across countries. In 2019 these ran from −42.4% **(Italy)** through −2.7% **(US, after the 2017 TCJA's immediate expensing)** to +23.9% **(Chile)** — a negative effective rate is a net *subsidy* to capital.
  - **Evidence and lineage.** Taxing *nominal* rather than real profits itself distorts investment (Feldstein, 1982). The user-cost elasticity of investment is estimated at about −0.66 (Cummins–Hubbard–Hassett, 1994), and the whole framework traces to **Hall–Jorgenson (1967)**.
  - **The policy hook.** This is exactly why governments use **investment tax credits** or **faster (accelerated) depreciation** — both lower the effective rate, cut the tax-adjusted user cost, and spur investment.

4.2.11 From the Desired Capital Stock to Investment

- We now know K^*. The next question is how the *actual* stock changes toward it, and that happens through **two opposing channels**.
- New capital $\Rightarrow$ **gross investment** (adds to the stock)
  - Gross investment is the new capital *added* this period — the flow that appears in GDP.
- Depreciation $\Rightarrow$ wears the stock **down**
  - Depreciation is the wear-out that *shrinks* the stock each period, pulling the actual capital in the opposite direction to gross investment.

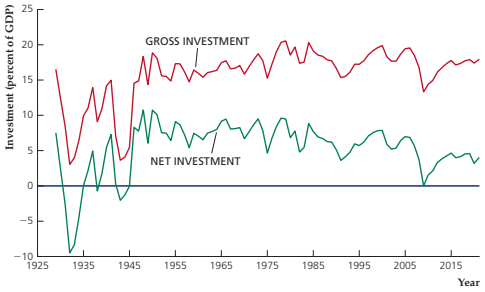
4.2.12 Net Investment

- **Net investment** = gross I − depreciation ⇒

$$K_{t+1} - K_t = I_t - dK_t \qquad (4.5)$$

- Net investment = **change** in the capital stock ⇒ $I >$ dep. ⇒ stock grows
  - Read (4.5) as a bathtub: gross investment I_t pours in, depreciation dK_t drains out, and the net change is the difference. If $I >$ depreciation the stock **grows**; if $I <$ depreciation it **shrinks**, even though gross investment is still positive.
- **One d hides enormous variation.** The single depreciation rate is an abstraction: BEA rates run from about **55%/yr** for prepackaged software and **31%/yr** for office equipment down to **~2.5%/yr** for office buildings and **~1%/yr** for houses. Since $uc = (r + d)p_K$, the d term *dominates* user cost for short-lived IT capital but is negligible for structures — so the same fall in r moves equipment and building investment very differently (Geerolf, ECON 102, from BEA).

4.2.13 Gross and Net Investment, 1929–2021



US gross and net investment as a share of GDP since 1929; the vertical gap between the two lines is depreciation.

Take-away. The gap between gross and net investment *is* depreciation; only *net* investment actually grows the capital stock.

- **Reading the figure.** Both series are plotted as a **percent of GDP**, and the distance between the gross and net lines is exactly depreciation — a visual of equation (4.5).
- **When net investment went negative.** During the **Great Depression** and **WWII (1941–45)**, net investment fell *below zero*: depreciation outran gross investment and the capital stock was actually **shrinking**. A striking reminder that an economy can disinvest.
- **A recent low.** Net investment was **very low in 2009–2011**, in the aftermath of the financial crisis — firms barely replaced worn-out capital, let alone expanded.

4.2.14 From the Desired Capital Stock to Investment (continued)

- Rewriting (4.5) gives $I_t = K_{t+1} - K_t + dK_t$
  - Just solve the net-investment identity (4.5) for **gross** investment I_t — no new economics yet, only algebra.
- Assume adjust in one period ⇒ $K^* = K_{t+1}$
  - **The one-period adjustment assumption.** With all available information, firms are taken to reach the desired stock *immediately* — the actual stock at the start of $t + 1$ equals K^*. This is a simplification; the lags slide walks it back.
- So $I_t = K^* - K_t + dK_t$

$$I_t = K^* - K_t + dK_t \qquad (4.6)$$

- **Equation (4.6) splits investment into two meaningful pieces:** the **gap to close** $(K^* - K_t)$, i.e. the net addition needed to reach the desired stock, plus **replacement** (dK_t) to cover what wore out. This decomposition is the whole payoff of the chapter so far.

4.2.15 Investment Has Two Parts

- **Net increase** ⇒ reach the desired stock $(K^* - K_t)$
  - This is the forward-looking piece: it carries *all* the K^* comparative statics (so it responds to r, MPK^f, and τ). When economists say investment is volatile, this is the term doing the moving.
- **Replacement** ⇒ cover depreciation (dK_t)
  - The replacement piece just makes good what wore out this period. It is tied to the existing stock, so it is comparatively **steady** — which is another reason the gap term, not replacement, drives the business cycle.
- **The same logic covers inventory and residential investment.** Business fixed capital is only one kind. Patricia, a car dealer deciding whether to raise her lot from 100 to 150 cars, weighs the extra sales commissions (the expected future marginal product) against interest on the floor-plan loan plus depreciation of unsold cars (the user cost) — she invests if benefit $\geq uc$. Houses obey the same rule, which is why the volatile housing cycle is first-order for investment (Abel Ch. 4).

4.2.16 Lags and Investment

- Some capital installs fast ⇒ other capital takes *years*
  - A laptop is in use the day it arrives; **factories and infrastructure** have long **build times**. So the one-period adjustment of (4.6) is a convenient fiction for the slowest kinds of capital.
- So adjustment to K^* ⇒ **spread over several years**
  - Real investment adjusts **gradually** toward K^*, not all at once. This is the basis of the "**Time to Build**" view of the cycle (Kydland–Prescott, 1982): the construction lag itself becomes a source of persistent, hump-shaped investment dynamics.

4.2.17 Investment and the Stock Market: Tobin's q

$$q = \frac{\text{market value of capital}}{\text{replacement cost of capital}}$$

- $q > 1$ ⇒ capital worth **more** than it costs ⇒ **invest**
- $q < 1$ ⇒ **do not** invest
  - **The definition, precisely.** q is the market value of a firm's capital (**equity plus debt**) divided by the **replacement cost** of its capital stock, from James Tobin's work.
  - **Reading the three cases.** If $q > 1$ the market values installed capital *above* the cost of building new, so firms have an incentive to **invest and expand**; if $q < 1$ capital is worth less than its replacement cost, so firms **disinvest** or let it depreciate; $q = 1$ is equilibrium, with no incentive to change the stock.

- **Why macroeconomists care.** The q theory **links financial markets to real investment** — stock prices rise → q rises → investment rises — and it is the same forward-looking logic as the user-cost rule, just read off asset prices instead of forecasts.
- **The testable version.** Under perfect competition and constant returns, **marginal q equals average q** (Hayashi, 1982), which is what makes q — an average, observable from market data — a valid stand-in for the marginal object theory wants.
- **Red flag: q alone predicts investment poorly.** Empirically q is a **weak sole predictor**; **cash flow** usually adds explanatory power (read as evidence of **financing constraints**), and **measurement error** in q (average vs. marginal) is a standing problem.
- **Ties back to lags.** Because of build lags, even when $q > 1$ the investment response is **delayed and gradual**, not instantaneous.

4.3 Goods Market Equilibrium

4.3.1 The Goods Market Equilibrium Condition

$$Y = C^d + I^d + G \tag{4.7}$$

- Equilibrium ⇒ output Y equals total desired spending $C^d + I^d + G$
  - Read (4.7) as a *clearing* condition, not an accounting truth: the goods market is in equilibrium only when what firms produce is exactly what households, firms, and government *want* to buy at the going interest rate.
- Unlike the *identity*, this condition **need not** always hold
  - **The common trap: confusing the identity with the equilibrium condition.** The identity $Y = C + I + G$ is true by definition in every period, because *unwanted* inventory is counted as part of measured I. The equilibrium condition (4.7) uses *desired* spending C^d, I^d, G — so it is a genuine economic restriction that can fail.
- Undesired output produced ⇒ unsold goods ⇒ **not** in equilibrium
  - When firms produce faster than buyers want, the surplus piles up as **unplanned inventory**. The identity still balances (that inventory is recorded as investment), yet the market is out of equilibrium. The force back to balance is quick — firms cut production or prices, and the real interest rate adjusts.
- Equivalently ⇒ desired national saving equals desired investment
  - **Why the two forms are the same.** Subtract $C^d + G$ from both sides of (4.7): the left becomes $Y - C^d - G$, which *is* desired national saving S^d, and the right becomes I^d. So saving-equals-investment is just (4.7) rearranged, not a separate assumption.

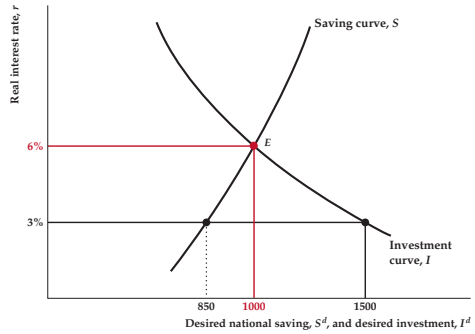
$$S^d = Y - C^d - G \quad \Longrightarrow \quad S^d = I^d \tag{4.8}$$

- **The one line to carry forward:** the **real interest rate r is the price that clears the goods market**. It moves until desired saving and desired investment are equal — exactly the role wages play in the labour market and the price level plays in the asset market.
- **Why r is the variable that moves (count the equations).** The closed economy is five relations

— $Y = C^d + I^d + G$, $C^d = C(Y - T)$, $I^d = I(r)$, $G = \bar{G}$, $T = \bar{T}$ — on top of $Y = \bar{Y}$ fixed by the supply side. Everything is pinned down *except* r, so r is the one free price that must adjust to make the resource constraint hold (Pérez, ECO3302).

- **Saving splits in two.** $S^d = \underbrace{(Y - T - C^d)}_{\text{private}} + \underbrace{(T - G)}_{\text{public}}$: national saving is private saving plus the government's budget balance. A deficit ($G > T$) is negative public saving that drags S^d down — the lever behind the fiscal experiments below (Pérez, ECO3302).

4.3.2 Goods Market Equilibrium: Plotting S^d vs. I^d



The saving curve slopes up (a higher r rewards waiting), the investment curve slopes down (a higher r raises the user cost of capital); they cross at the equilibrium real interest rate.

Take-away. The master diagram of the chapter: the real interest rate adjusts until desired saving equals desired investment.

- Equilibrium at point **E** where $S^d = I^d$ — here $r = 6\%$ and $S^d = I^d = 1000$
  - **The teaching point:** everything else in the section is this one diagram with a single curve shifted — read the equilibrium straight off where S^d meets I^d.
- Off equilibrium, r moves to close the gap between S^d and I^d
  - **Worked disequilibrium.** At $r = 3\%$, desired investment $I^d = 1500$ exceeds desired saving $S^d = 850$ — an **excess demand** of 650 for goods. Would-be borrowers compete for scarce saving and bid the rate **up**, $3\% \to 6\%$, until S^d rises to 1000, I^d falls to 1000, and the market lands back at **E**.
  - The two curves have the shapes we derived earlier: saving rises in r (the reward for postponing consumption), investment falls in r (a higher rate lifts the user cost). Where they cross is the *only* rate at which plans are mutually consistent.

4.3.3 Components of Aggregate Demand for Goods (An Example)

Variable	$r = 3\%$	$r = 6\%$
Output, Y	4500	4500
Desired consumption, C^d	2150	2000
Desired investment, I^d	1500	1000
Government purchases, G	1500	1500
Desired national saving, $S^d = Y - C^d - G$	850	1000
Aggregate demand, $C^d + I^d + G$	5150	4500

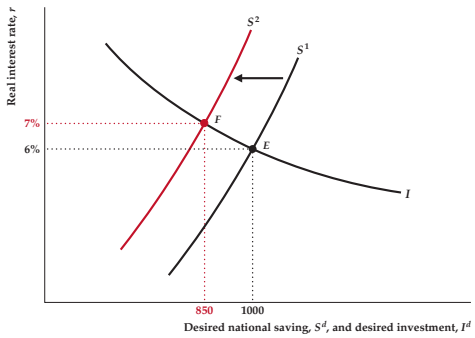
- **The $r = 6\%$ column is the equilibrium:** aggregate demand $C^d + I^d + G = 2000 + 1000 + 1500 = 4500$ exactly matches output $Y = 4500$, and equivalently $S^d = I^d = 1000$. **Both** equilibrium conditions — (4.7) $Y = C^d + I^d + G$ and (4.8) $S^d = I^d$ — hold together at that one rate.

- At $r = 3\%$ the economy is in **excess demand**, not equilibrium
  - **Read the first column.** A lower rate makes consumers save less and firms invest more, so C^d (2150) and I^d (1500) are both higher than at 6%. Demand = 2150 + 1500 + 1500 = 5150 overshoots supply $Y = 4500$; equivalently $I^d = 1500 > S^d = 850$. Goods are over-demanded by 650.
- Raising r to 6% removes the excess demand — two reads of one adjustment
  - *Saving = investment, Eq. (4.8):* a higher r makes saving more attractive and investment more costly, so S^d **rises** (850 → 1000) and I^d **falls** (1500 → 1000) until they meet.
  - *Output = demand, Eq. (4.7):* a higher r **cuts both** C^d and I^d, so aggregate demand falls (5150 → 4500) until it meets supply.
  - **Why the two agree — a frequent confusion.** Since $S^d = Y - C^d - G$, *cutting* C^d *is the same thing as raising* S^d. The "demand falls" story and the "saving rises" story describe one adjustment from two angles; they are not in conflict.

4.3.4 Worked Example: Solving for the Market-Clearing Rate

- A fully specified version of the clearing mechanism (Pérez, ECO3302). Take $Y = 1000$, $G = 180$, $T = 300$, $C^d = 160 + 0.7(Y - T)$, and $I^d = 400 - 25r$.
- **Find desired saving.** $C^d = 160 + 0.7(1000 - 300) = 650$, so $S^d = Y - C^d - G = 1000 - 650 - 180 = 170$.
  - Equivalently, private saving = $Y - T - C^d = 50$ plus public saving = $T - G = 120$, giving $S^d = 170$.
- **Set $S^d = I^d$ and solve.** $170 = 400 - 25r \Rightarrow r = 9.2\%$ — the market-clearing real rate.
- **Now cut taxes to $T = 200$ (with G fixed).** Disposable income rises, so C^d rises by MPC × ΔT = $0.7 \times 100 = 70$, to 720; desired saving falls to $S^d = 1000 - 720 - 180 = 100$.
  - New clearing rate: $100 = 400 - 25r \Rightarrow r = 12\%$. The deficit-financed tax cut lowered national saving and pushed r up from 9.2% to 12% — crowding out, with the **MPC** setting the size of the move.

4.3.5 A Decline in Desired Saving



Lower desired saving shifts the saving curve left ($S_1 \to S_2$); equilibrium slides up the investment curve from **E** to **F**, raising r and cutting investment.

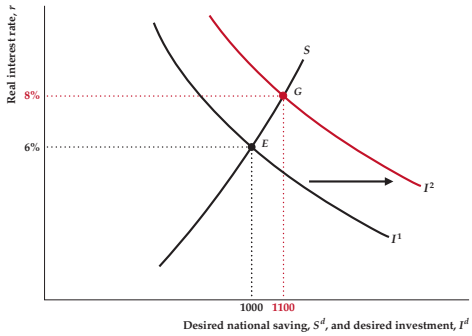
Take-away. Less desired saving ⇒ a higher r *and* less investment — this is crowding out.

- **Same numerical example continues.** The starting point **E** is the earlier equilibrium — $r = 6\%$, $S^d = I^d = 1000$, exactly the cross from the previous figure.
- Desired saving falls (e.g. a temporary $G \uparrow$) ⇒ saving curve shifts **left**

- A temporary rise in government purchases is the textbook trigger: households cut desired saving to help pay for it, so the whole saving curve moves left ($S_1 \to S_2$) at every interest rate.

- New equilibrium **F**: r rises 6% → 7% and S, I fall 1000 → 850
  - **This is crowding out.** With less saving to fund it, investment is squeezed as the rate climbs. The economics is the single arrow: less saving ⇒ higher r ⇒ less investment.
  - **The true cost falls on the future, not today.** Lower investment means a smaller **future capital stock**, so the economy is less productive later. In effect, *future generations* bear the burden of today's deficit — the headline lesson of the saving-shift experiment.

4.3.6 An Increase in Desired Investment



Higher desired investment shifts the investment curve right ($I_1 \to I_2$); equilibrium moves up the saving curve from **E** to **G**, so both r and the common level of S, I rise.

Take-away. An investment boom raises r *and* the quantity traded, because the economy climbs a fixed, upward-sloping saving curve.

- **Same example, same starting point E** ($r = 6\%$, $S^d = I^d = 1000$) — but now the *investment* curve shifts, not saving.
- Desired investment rises (e.g. an invention raising MPK^f) ⇒ investment curve shifts **right**
  - A new technology or invention lifts the future marginal product of capital MPK^f, so firms want more capital at every interest rate; the investment curve moves right ($I_1 \to I_2$).
- New equilibrium **G**: r rises 6% → 8% and S, I rise 1000 → 1100
  - **Contrast with the saving-fall case.** Here **both** saving and investment end up *higher*, because the economy slides up the fixed saving curve: a higher r coaxes savers to save more, and that extra saving funds the extra investment. So an investment boom raises r *and* the quantity traded, whereas a saving decline raised r but *cut* the quantity.
  - **What the upward-sloping saving curve buys you.** If desired saving were *independent* of r (a vertical S curve), a rightward investment shift would raise r but leave equilibrium investment *unchanged* — saving alone would pin the quantity down. Both r and I rise only because S slopes up (Pérez, ECO3302).
- **"Crowded in," not always out.** Infrastructure spending raises G *and* the future MPK^f, so it shifts *both* curves — S left (future taxes) and I right (higher MPK^f). r rises, but investment can end up *higher*: the government has then *crowded in* private investment, and it is consumption that gets crowded out instead. The reflexive "$G \uparrow \Rightarrow I \downarrow$" is not a law.
- **Policy question — should tax policy chase more saving and investment?** The chapter's levers do work: IRAs raise the after-tax return to saving, and business tax breaks cut the after-tax user cost of capital. Two cautions follow. First, is investment actually *too low*? More saving means less consumption *today*, which households may not want. Second, over-encouraging capital can leave an *inefficiently large* stock — the late-1980s glut of commercial office space, partly

tax-driven, is the cautionary case. A sound case for intervention needs a genuine **externality**, not merely a wish for faster growth.

4.3.7 A Different Lens: What If Output Isn't Fixed?

Beyond Abel's model — a different regime, kept separate

Abel's §4-3 holds output at full employment $\bar{Y}$ and lets the real interest rate clear the market. Drop that assumption — as a short-run, slack-economy model does — and two of its results flip. These belong to a *different regime*, not a correction of the above; keep them apart from the Abel derivation.

- **The paradox of thrift.** If output is *demand-determined* (Keynes, *General Theory*, 1936, Ch. 7), a rise in desired saving cuts consumption → cuts income → can leave actual saving *unchanged or lower*: "every such attempt to save more… necessarily defeats itself." Crucially, both Keynes (Ch. 24) and Samuelson (1948) say this holds *only* below full employment — "if we were at full employment… the old-fashioned doctrine of thrift would be absolutely correct." The paradox is precisely the regime Abel assumes away.
- **Crowding in, empirically.** Abel predicts a smaller deficit lowers r and crowds investment *in*; tax-based austerity instead shows a long-run *fall* in investment (Alesina–Favero–Giavazzi, 2019; Romer & Romer, 2010), with C, I, and Y moving *together* — a tension with the full-employment comparative statics worth flagging.
- **Bottom line:** the two frameworks *agree* at full employment and diverge in a slack economy. Naming the assumption ($Y = \bar{Y}$) makes it visible rather than silent.

4.4 Appendix: The Business Cycle

- **Why this sits here.** Investment is the most volatile piece of GDP, so the chapter's machinery is also the natural entry point to the *fluctuations* that Part 3 studies in depth. This appendix just fixes the vocabulary.
- **Business cycle** ⇒ recurring **expansions** and **contractions** of real GDP around its long-run **trend** (potential output)
  - The cycle is measured as deviations *from* trend, not the level of GDP itself: an economy growing at 2% is still "in recession" if it is below the path it normally follows.
- **Peak / boom** then **trough / bust** ⇒ turning points of the cycle
  - A **peak** is the high turning point where expansion ends; a **trough** is the low turning point where contraction ends and recovery begins. In the US these dates are called by the NBER, not by a fixed "two quarters of falling GDP" rule.
- **Contraction** (recession) then **expansion** (recovery) ⇒ the two phases
  - **Who built the idea.** Burns & Mitchell (1946, NBER) first defined and measured cycles empirically; Schumpeter tied them to waves of **innovation** and "creative destruction"; Kydland & Prescott (1982) launched **real-business-cycle** theory, treating cycles as the efficient response to productivity shocks.

4.5 Taking Stock

4.5.1 Putting it all together

- **The chapter in four lines:** desired saving $S^d = Y - C^d - G$; desired capital where $MPK^f = uc$ with $uc = (r + d)p_K$; investment $I_t = K^* - K_t + dK_t$; equilibrium $S^d = I^d$. Reproduce these four and you have Ch 4.
- **Consumption and saving**: one decision, $S^d = Y - C^d - G - C^d$ **rises** with current income (MPC between 0 and 1), expected future income, and wealth, and the net effect of a higher r is a **small** rise in saving
- **Desired capital stock**: $MPK^f = uc$, i.e. $uc = (r + d)p_K$ — a lower r, d, or p_K, a higher expected MPK^f, or a lower effective tax rate all **raise** K^*
- **Investment**: $I_t = K^* - K_t + dK_t$ — close the gap to K^* **plus** replace depreciation, with build lags and Tobin's q shaping the timing
- **Goods-market equilibrium**: $Y = C^d + I^d + G \Leftrightarrow S^d = I^d$ — the **real interest rate** is the price that clears the market
  - The hinge is the S–I **diagram**: desired saving slopes **up** and desired investment slopes **down** in r, so they cross at one market-clearing rate; everything before it is machinery for reading that one picture.
- **The deepest idea — the real interest rate reconciles two independent plans.** Households decide how much to save and firms decide how much to invest for entirely separate reasons, yet in a closed economy the two must be equal. It is r that adjusts until desired national saving equals desired investment — the goods market's single price.
- **A caveat worth carrying:** all of this assumes output is fixed at full employment $\bar{Y}$. Drop that (a slack economy) and two results flip — the paradox of thrift and crowding-in — so always name the assumption before applying the S–I logic.

4.5.2 Building blocks: how the pieces connect

Consumption–saving decision → Desired saving $S^d(r)$ (up) • **Desired capital** K^* → **Desired investment** $I^d(r)$ (down) → **Goods-market equilibrium** $S^d = I^d$ → **Market-clearing** r

- **The flow is the chapter order:** the saving side (how C^d and S^d are set) and the investment side (how K^* and I^d are set) are built separately, then brought together where $S^d = I^d$ pins down the real interest rate.
- The goods market is the **second leg** of the Ch 9 three-market model (labour · **goods** · assets)
  - Its job in that model: it supplies the **IS** relation — the goods-market-clearing combinations of output and the real interest rate.
- It pins down the **real** side — output and the real interest rate — given full-employment $\bar{Y}$ from the labour market of Ch 3
- **What's next:** Ch 3 gave the labour market and $\bar{Y}$; this chapter built the **goods market** (the S–I / IS leg); Ch 7 adds the **asset market**; Ch 9 stops passing r and P between markets one at a time and solves all three **together** (**IS–LM–FE**).

Economists behind this chapter

David Ricardo (1772–1823, British) — classical economist whose argument that government borrowing and taxation can be equivalent is the origin of the *Ricardian equivalence* proposition.

Irving Fisher (1867–1947, American) — *The Theory of Interest* (1930); the two-period model of intertemporal consumption choice and the real interest rate as the price of present versus future consumption.

Joseph Schumpeter (1883–1950, Austrian) — innovation-driven business cycles and "creative destruction."

John Maynard Keynes (1883–1946, British) — *The General Theory* (1936); the consumption function $C = c_0 + c_y Y$ and the marginal propensity to consume.

Wesley C. Mitchell (1874–1948, American) & **Arthur F. Burns** (1904–1987, American) — *Measuring Business Cycles* (NBER, 1946); the empirical definition and dating of cycles.

Milton Friedman (1912–2006, American) — the *permanent-income hypothesis* (1957): consumption tracks permanent, not current, income; Nobel 1976.

James Tobin (1918–2002, American) — the q theory of investment linking the stock market to capital spending; Nobel 1981.

Franco Modigliani (1918–2003, Italian-American) — the *life-cycle hypothesis* of consumption and saving; Nobel 1985.

Dale W. Jorgenson (1933–2022, American) — the neoclassical theory of investment and the *user cost of capital* (1963).

Martin Feldstein (1939–2019, American) — how taxes and inflation distort saving and investment through the user cost of capital.

Edward C. Prescott (1940–2022, American) & **Finn E. Kydland** (b. 1943, Norwegian) — "Time to Build and Aggregate Fluctuations" (1982), the foundation of *real-business-cycle* theory; Nobel 2004.

Robert E. Hall (b. 1943, American) — Hall–Jorgenson (1967) on tax policy and investment, and the random-walk theory of consumption (1978).

Robert J. Barro (b. 1944, American) — "Are Government Bonds Net Wealth?" (1974), the modern formalisation of Ricardian equivalence.

Fumio Hayashi (b. 1952, Japanese) — showed when marginal q equals average q (1982), making Tobin's q empirically usable.

References

Core texts (Abel is the uncited spine):

Abel, A. B., Bernanke, B. S., & Croushore, D. *Macroeconomics* (11th ed.). Pearson.

Mankiw, N. G. (2023). *Macroeconomics* (12th ed.). Worth.

Geerolf, F. *ECON 102* lecture notes (UCLA): L02 Consumption, L03 Investment, L04 Paradox of Thrift.

Pérez, L. *ECO3302* lecture notes: Lect. 4–5 (national income, the goods market).

Works cited in the notes

Agarwal, S., Liu, C., & Souleles, N. S. (2007). The reaction of consumer spending and debt to tax rebates — evidence from consumer credit data. *Journal of Political Economy*, 115(6), 986–1019.

Ahmed, S. (1986). Temporary and permanent government spending in an open economy. *Journal of Monetary Economics*, 17(2), 197–224.

Alesina, A., Favero, C., & Giavazzi, F. (2019). *Austerity: When It Works and When It Doesn't*. Princeton University Press.

Barro, R. J. (1974). Are government bonds net wealth? *Journal of Political Economy*, 82(6), 1095–1117.

Bernheim, B. D. (1987). Ricardian equivalence: An evaluation of theory and evidence. *NBER Macroeconomics Annual*, 2, 263–304.

Burns, A. F., & Mitchell, W. C. (1946). *Measuring Business Cycles*. NBER.

Cummins, J. G., Hubbard, R. G., & Hassett, K. A. (1994). A reconsideration of investment behavior using tax reforms as natural experiments. *Brookings Papers on Economic Activity*, 1994(2), 1–74.

Feldstein, M. (1982). Inflation, tax rules, and investment: Some econometric evidence. *Econometrica*, 50(4), 825–862.

Fisher, I. (1930). *The Theory of Interest*. Macmillan.

Friedman, M. (1957). *A Theory of the Consumption Function*. Princeton University Press.

Hall, R. E., & Jorgenson, D. W. (1967). Tax policy and investment behavior. *AER*, 57(3), 391–414.

Hayashi, F. (1982). Tobin's marginal q and average q: A neoclassical interpretation. *Econometrica*, 50(1), 213–224.

Jorgenson, D. W. (1963). Capital theory and investment behavior. *AER*, 53(2), 247–259.

Keynes, J. M. (1936). *The General Theory of Employment, Interest and Money*. Macmillan.

Kydland, F. E., & Prescott, E. C. (1982). Time to build and aggregate fluctuations. *Econometrica*, 50(6), 1345–1370.

Modigliani, F. (1986). Life cycle, individual thrift, and the wealth of nations. *AER*, 76(3), 297–313.

Modigliani, F., & Brumberg, R. (1954). Utility analysis and the consumption function. In K. Kurihara (Ed.), *Post-Keynesian Economics*. Rutgers University Press.

O'Driscoll, G. P. (1977). The Ricardian nonequivalence theorem. *Journal of Political Economy*, 85(1), 207–210.

Parker, J. A., Souleles, N. S., Johnson, D. S., & McClelland, R. (2013). Consumer spending and the economic stimulus payments of 2008. *AER*, 103(6), 2530–2553.

Romer, C. D., & Romer, D. H. (2010). The macroeconomic effects of tax changes. *AER*, 100(3), 763–801.

Saez, E., & Zucman, G. (2016). Wealth inequality in the United States since 1913. *Quarterly Journal of Economics*, 131(2), 519–578.

Samuelson, P. A. (1948). *Economics: An Introductory Analysis*. McGraw-Hill.

Shapiro, M. D., & Slemrod, J. (2003). Consumer response to tax rebates. *AER*, 93(1), 381–396.

Tobin, J. (1969). A general equilibrium approach to monetary theory. *Journal of Money, Credit and Banking*, 1(1), 15–29.

***** END *****

To get a great answer, you must first ask a good question!

— Anonymous

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